Skills Summary

Derivatives: Power, Product, and Quotient Rules

Use this reference while completing your worksheet or when studying.

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1. Power rule (rewrite first)

Rule. $\dfrac{d}{dx}x^n = nx^{n-1}$ for every real n, and $\dfrac{d}{dx}\left[c\,f(x)\right] = c\,f'(x)$; the derivative of a constant is 0.

Rewrite before you differentiate: $\sqrt{x} = x^{1/2}$, $\sqrt[3]{x} = x^{1/3}$, $\dfrac{1}{x^n} = x^{-n}$. The power rule cannot see a radical or a reciprocal until it is a power.

Example: Differentiate $y = 3x^4 - \dfrac{5}{x}$.

Step 1. The second term is not written as a power, so rewrite it as $-5x^{-1}$ and then every term is power-rule ready.

Step 2. $3x^4 \to 12x^3$; $-5x^{-1} \to +5x^{-2}$ (the exponent -1 comes out front and drops to -2).

$\Rightarrow \quad \boldsymbol{y' = 12x^3 + \dfrac{5}{x^2}}$

Try it: Differentiate $y = 2\sqrt{x} + x^3$.

check: $\boldsymbol{y' = \dfrac{1}{\sqrt{x}} + 3x^2}$

2. Product rule

Rule. $(uv)' = u'v + uv'$ — "derivative of the first times the second, plus the first times the derivative of the second".

The derivative of a product is **never** $u'v'$. Label u, v, u', v' before substituting, and you will not lose a term.

Example: Differentiate $y = (x+4)(x^2-2)$.

Step 1. Two factors, each of which is easy to differentiate on its own — that is the product rule's signature.

Step 2. $u = x+4$, $u' = 1$; $v = x^2-2$, $v' = 2x$. Then $y' = 1(x^2-2) + (x+4)(2x) = x^2 - 2 + 2x^2 + 8x$.

$\Rightarrow \quad \boldsymbol{y' = 3x^2 + 8x - 2}$

Try it: Differentiate $y = (3x-1)(x^2+5)$.

check: $\boldsymbol{y' = 9x^2 - 2x + 15}$

3. Quotient rule

Rule. $\left(\dfrac{u}{v}\right)' = \dfrac{u'v - uv'}{v^2}$ — "low · d-high minus high · d-low, all over low squared".
The subtraction makes the **order matter**: swapping the two products flips the sign of the whole answer. Leave v^2 factored; expanding it hides the structure and helps nothing.

Example: Differentiate $f(x) = \dfrac{x+2}{x-3}$.
Step 1. A single fraction with a variable in the denominator, and no easy rewrite as a power — use the quotient rule.
Step 2. $u = x+2$, $u' = 1$; $v = x-3$, $v' = 1$. Numerator: $1(x-3)-(x+2)(1) = x-3-x-2 = -5$.

$$\Rightarrow \quad \boldsymbol{f'(x) = \frac{-5}{(x-3)^2}}$$

Try it: Differentiate $f(x) = \dfrac{2x}{x+1}$.

check: $\boldsymbol{f'(x) = \dfrac{2}{(x+1)^2}}$

4. Where the rules come from: the limit definition

Definition. $f'(x) = \lim\limits_{h\to 0} \dfrac{f(x+h)-f(x)}{h}$.
The plan is always the same: expand $f(x+h)$, subtract $f(x)$ so the constant terms cancel, factor h out of what is left, cancel it, and only then let $h \to 0$.

Example: Find $f'(x)$ from the definition for $f(x) = x^2$.
Step 1. Substituting $h = 0$ straight away gives $0/0$, so the h in the denominator has to be cancelled first.
Step 2. Numerator: $(x+h)^2 - x^2 = 2xh + h^2 = h(2x+h)$. Cancel the h to get $2x+h$, then let $h \to 0$.

$$\Rightarrow \quad \boldsymbol{f'(x) = 2x}$$

Try it: Find $f'(x)$ from the definition for $f(x) = 4x^2$.

check: $\boldsymbol{f'(x) = 8x}$